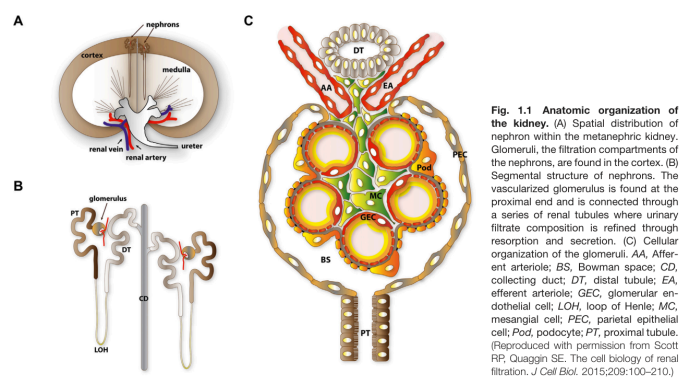
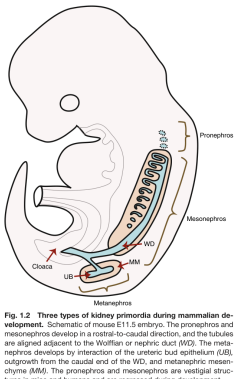
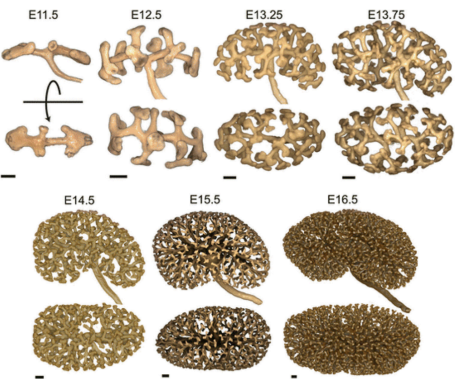




Fig_1_1



Fig_1_2



Fig_1_3

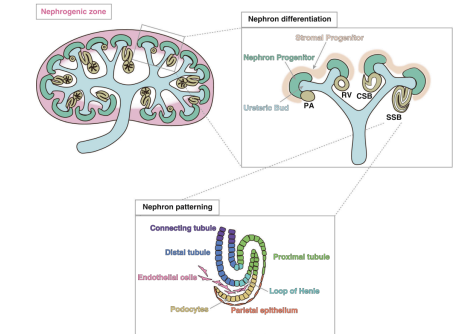
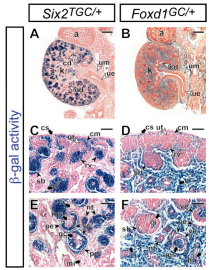
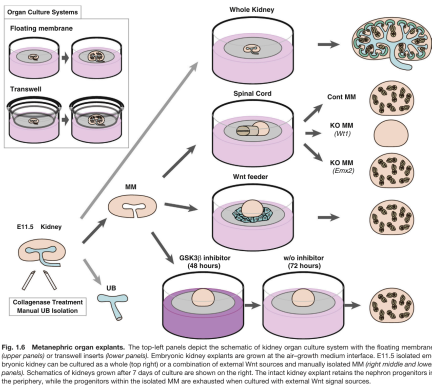


Fig. 1.4 Overview of nephrogenesis. Reciprocal interaction among the nephron progenitor, stromal progenitor, and ureteric bud within the nephrogenic zone enables continuous branching of the ureteric epithelium, self-renewal of nephron progenitor, and differentiation into epithelialized nephron structures. Successive proximo-distal patterning within the nascent nephron structure results in the formation of an S-shaped body with distinct gene expression patterning coding for the future nephron domains. CSB, comma-shaped body; PA, pretubular aggregate; RV, renal vesicle; SSB, S-shaped body.

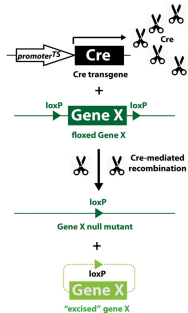
Fig_1_4



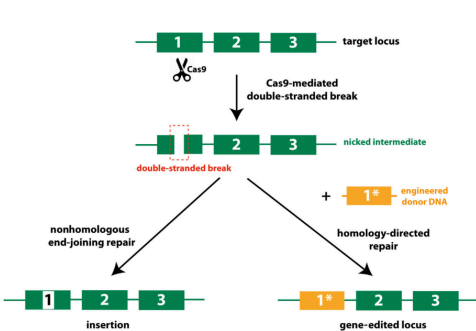
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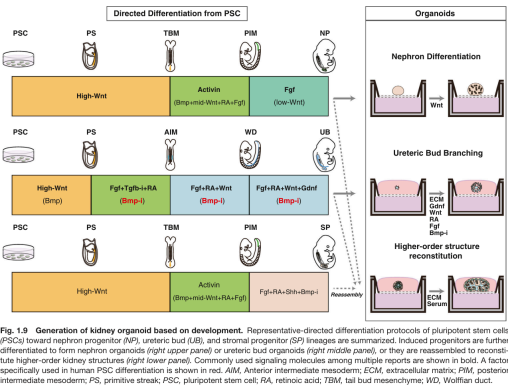
Fig_1_6



Fig_1_7



Fig_1_8



Fig_1_9

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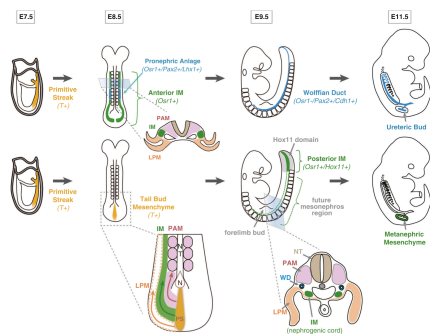


Fig. 1.10 Developmental origin of metanephric kidney progenitors. Model for lineage segregation of the ureteric bud (UB) and metanephric mesenchyme (MM). The UB differentiates from E7.5 T-positive primitive streak through the pronephric anlage, a part of anterior intermediate mesoderm (IM) at E8.5. The MM originates from the E7.5 T3-positive primitive streak, then stays in the T-positive tail bud mesenchyme until E8.5 and finally differentiates into posterior IM at E9.5. The section image shows the E8.5 and E9.5 embryonic stages is shown on the side panels. The magnified image of the tailbud mesenchyme region shows the cell migration from the primitive streak. *LPK*, Lateral plate mesoderm; *PMK*, paraxial mesoderm; *N*, notochord; *NT*, neural tube; *W*, Wolffian duct.

Fig 1 10

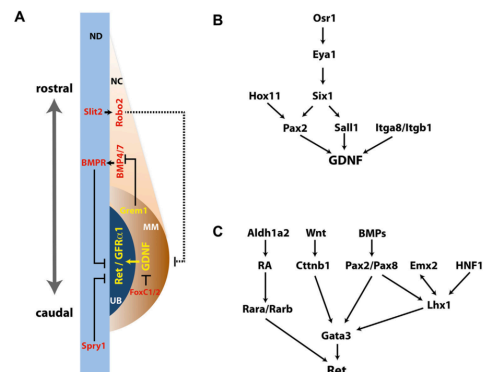


Fig. 1.11 Genetic interactions during early metanephric kidney development. (A) Regulatory interactions control the strategically localized expression of glial cell-derived neurotrophic factor (GDNF) and Ret and the subsequent induction of the ureteric bud. The anterior part of GDNF expression is restricted by Foxc1/2 and Slit2/Robo2 signaling. Spry1 suppresses the postreceptor activity of Ret. BMP4/7-BMPR signaling inhibits the response to GDNF, an effect counteracted by Grem1. The genetic regulatory network controls the expression of (B) GDNF and (C) Ret. BMPR, Bone morphogenetic protein receptor; MM, metanephric mesenchyme; NP, nephrogenic cord; NQ, neuregulin duct; UR, ureteric bud.

Fig 1 11

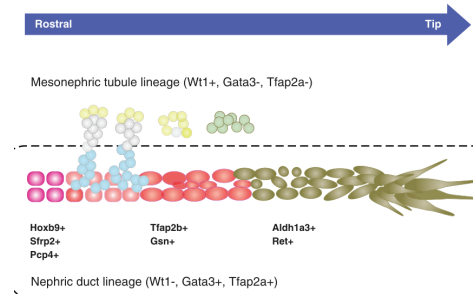


Fig. 1.12 Spatial organization of nephric duct progenitors. Spatial development of the nephric duct. Representative genes expressed in each region are shown. (Modified with permission from Sanchez-Ferraz O, Pacis A, Sotiropoulou M, et al. A coordinated progression of progenitor cell states initiate urinary tract development. *Nat Commun.* 2021;12:2627.)

Fig 1 12

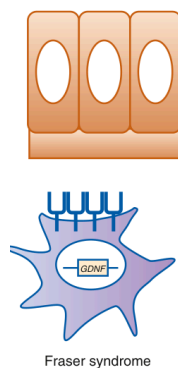


Fig. 1.13 Molecular model of renal defect in Fraser syndrome. (A) Adhesion to the ureteric bud (UB) epithelium positively regulates the expression of glial cell-derived neurotrophic factor (GDNF) by the metanephric mesenchyme (MM). Adhesion and GDNF expression are impaired in the absence of (B) nephroneurin (expressed by the UB), (C) $\alpha 8 \beta 1$ integrin (expressed by the MM), and (D) or the Fras1/Frem1/Frem2 complex. Fras1, Frem1, and Frem2 are implicated in Fraser syndrome and are believed to coordinately anchor nephroneurin to the UB basement membrane and stabilize the conjugation with $\alpha 8 \beta 1$ integrin. (Modified from Kiyosumi, Takeichi M, Nakano I, et al. Basement membrane assembly of the integrin $\alpha 8 \beta 1$ ligand nephroneurin requires Fraser syndrome-associated proteins. *J Cell Biol*, 2012;197:677–689.)

Fig 1 13

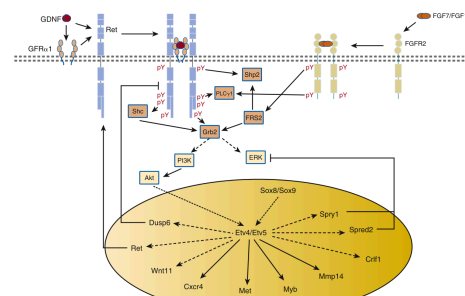


Fig. 1.14 Ret signaling pathway. Ret is activated and becomes autophosphorylated on intracellular tyrosine residues upon association with glial cell-derived neurotrophic factor (GDNF) and GFR1 α . Signaling molecules such as Grb2, Shc, FRS2, PLC γ , and Shp2 bind directly to the phosphorylated tyrosine residues within the intracellular domain of Ret. Recruitment of Shc, FRS2, and PLC γ leads to activation of extracellular signal-regulated kinase (ERK) and PI3K/Akt pathways. GDNF Ret signaling leads to specific activation of a host of genes, some of which are strongly dependent on the upregulation of the transcription factors E12 and E12-63 (red arrows). E12-63 itself requires activation of the PI3K/Akt but not the ERK pathway. Smad3 and Smad5 are believed to act in parallel to reinforce transcriptional responses to GDNF/Ret engagement. Some of these pathways are shared with the FGF7/10-FGFR2 receptor signaling system. Spry1 and Spry2 negatively regulate Ret signaling, whereas Dusp6 likely mitigates the dephosphorylation of the Ret receptor, thus acting as part of a negative feedback regulatory loop.

Fig 1 14

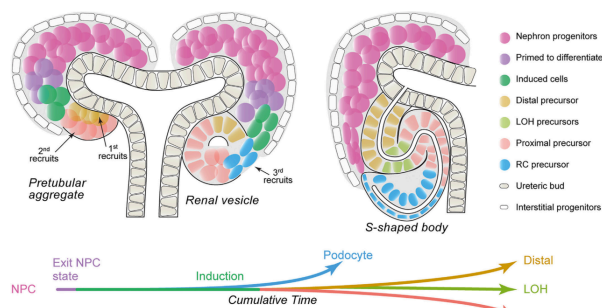


Fig. 1.16 Gradual recruitment of progenitor cells during nephrogenesis. Early recruited nephron progenitors clustering around the ureteric bud give rise to pretubular aggregates. New progenitors continue to be recruited and incorporate into the proximal end of the renal vesicle. Distal, medial, and proximal domains are established by the S-shaped body stage. *LOH*, Loop of Henle; *NPC*, nephron progenitor cell; *RC*, renal corpuscle. (Adapted from Lindström NO, De Sena Brandine G, Tran T, et al. Progressive recruitment of mesenchymal progenitors reveals a time-dependent process of cell fate acquisition in mouse and human nephrogenesis. *Dev Cell*. 2018;45:651–660).

Fig_1_16

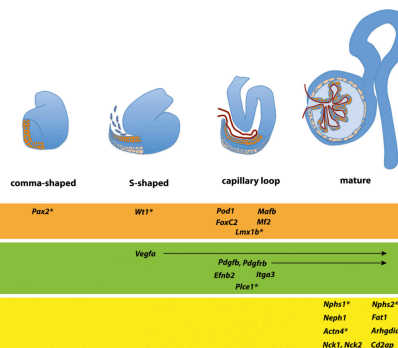


Fig. 1.17 Molecular basis of glomerular development. Key factors are shown along with the time point where major effects were observed in knockout or transgenic mouse studies. Many factors play roles at more than one time point. Genes identified as mutated in patients with glomerular disease are marked by asterisks.

Fig_1_17

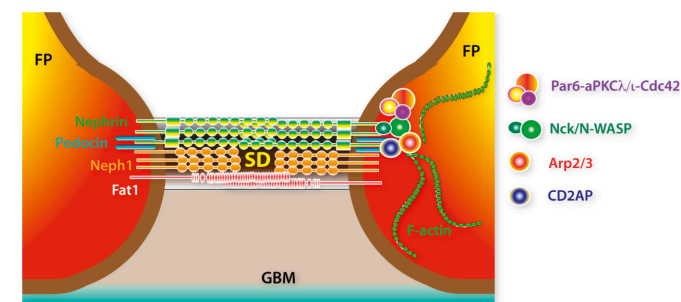


Fig. 1.18 Structural overview of the slit diaphragm (SD). An oversimplified diagram depicting the major adhesion receptors comprising the SD and how they are possibly integrated with the actin cytoskeleton of podocyte foot processes. *FP*, Podocyte foot process; *GBM*, glomerular basement membrane. (Adapted from Scott RP, Quaggin SE. Formation and maintenance of a functional glomerulus. In Little MH (ed.). *Kidney Development, Disease, Repair and Regeneration*. San Diego: Academic Press; 2016.)

Fig_1_18

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Table_e1_1_part1

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Table e1.1 Summary of Knockout and Transgenic Models for Kidney Development (Cont'd)			
Gene Mutation or Knockout	Phenotype	Associated Human Diseases	References
<i>Tcp2l1</i>	Loss of CD intercalated cells		340
<i>Tjp1</i> (ZO-1)	Podocyte effacement and proteinuria		713
<i>Tp63</i> (TP63)	Loss of CD intercalated cells	ADULT (acro-dermato-ungual-lacrimar-tooth) syndrome; limb-mammary syndrome	714
<i>Tpce6</i>	Protected from angiotensin-mediated or proteinuria or complement-dependent glomerular injury (null mutation); podocyte foot process effacement and proteinuria (transgenic overexpression in the podocyte lineage)	SRNS, FSGS	715,716,717,718, 713,719,720
<i>Vangl2</i>	Immature and poorly branched glomerular tuft	Neural tube defects	355
<i>Vegfa</i>	Endotheliosis, disruption of glomerular filtration barrier formation, nephrotic syndrome (podocyte-selective); peritubular capillary rarefaction and polycythemia (tubule-specific)		426,386,437,722
<i>Wt1</i>	Glomerulonephritis (podocyte-selective)	Von Hippel-Lindau syndrome	449
<i>Wasl</i> (N-wasp)	Podocyte effacement, proteinuria		723
<i>Wnt7b</i>	Impaired development medullary microvasculature		724
<i>Wnt11</i>	Glomerular cysts		725

ADPKD, Autosomal dominant polycystic kidney disease; ARPKD, autosomal recessive polycystic kidney disease; CAKUT, congenital anomalies of the kidney and urinary tract; CD, collecting duct; CM, cap mesenchyme; FSGS, focal segmental glomerulosclerosis; GBM, glomerular basement membrane; GDNF, glial cell-derived neurotrophic factor; MM, metanephric mesenchyme; SRNS, steroid-resistant nephrotic syndrome; UB, ureteric bud; VUR, vesicoureteral reflux.

Table_e1_1_part10